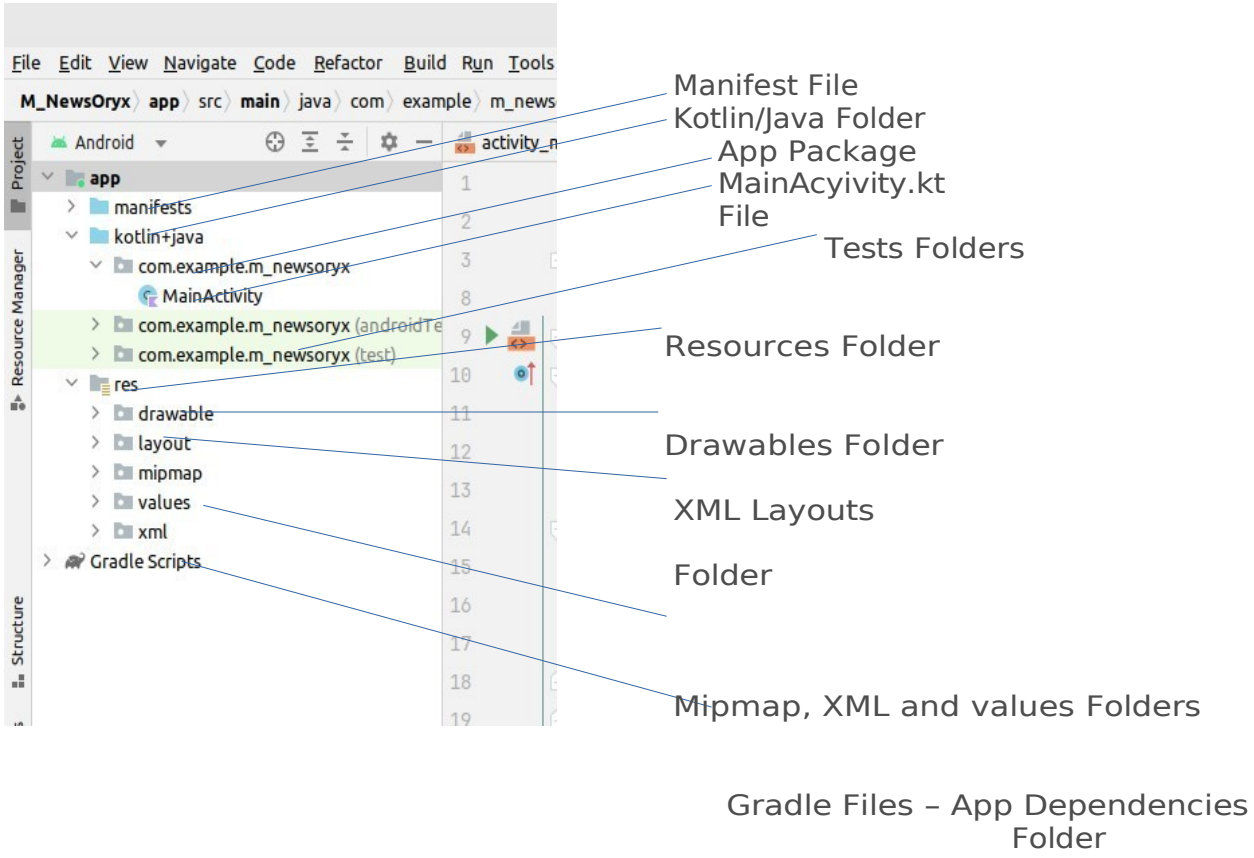
Android App Folder Structure

To implement android apps, Android Studio is the official IDE (Integrated Development Environment) which is freely provided by Google for android app development.

Once we setup android development environment using android studio and if we create a sample application using android studio, our project folder structure looks as shown below. In case if you are not aware of creating an application using an android studio please check PDF 002 Android Hello World App on How to Create Your First App.

Below We Look at a General App Structure For Android and Any Application, We use a sample app to Explain this.

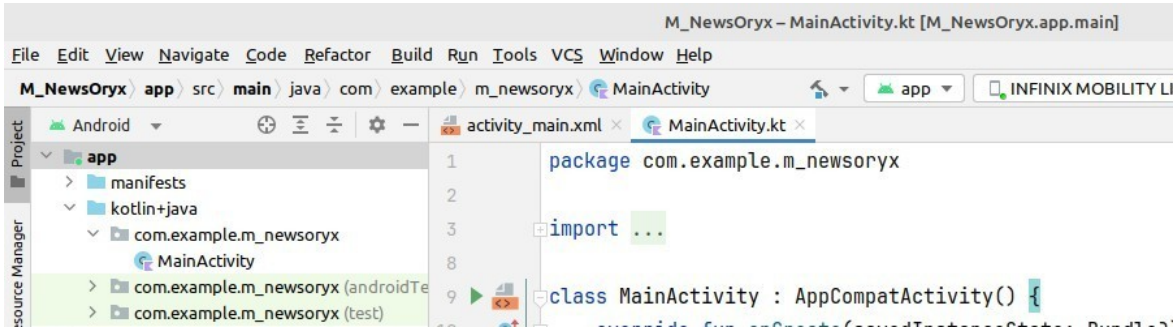


The Android project structure on the disk might be differs from the above representation. To see the actual file structure of the project, select **Project** from the **Project** dropdown instead of **Android**.

The android app project will contain different types of app modules, source code files, and resource files. We will explore all the folders and files in the android app.

Java/Kotlin Folder

This folder will contain all the Kotlin source code (**.kt**) files which we'll create during the application development, including JUnit test code. Whenever we create any new project/application, by default the class file **MainActivity.kt** create automatically under the package name "**com.example.m_newsoryx**" like as shown below.



res (Resources) Folder

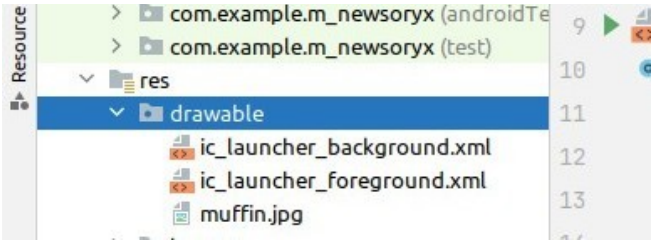
It's an important folder that will contain all non-code resources, such as bitmap images, UI strings, XML layouts like as shown below.



The res (**Resources**) will contain a different type of folders that are

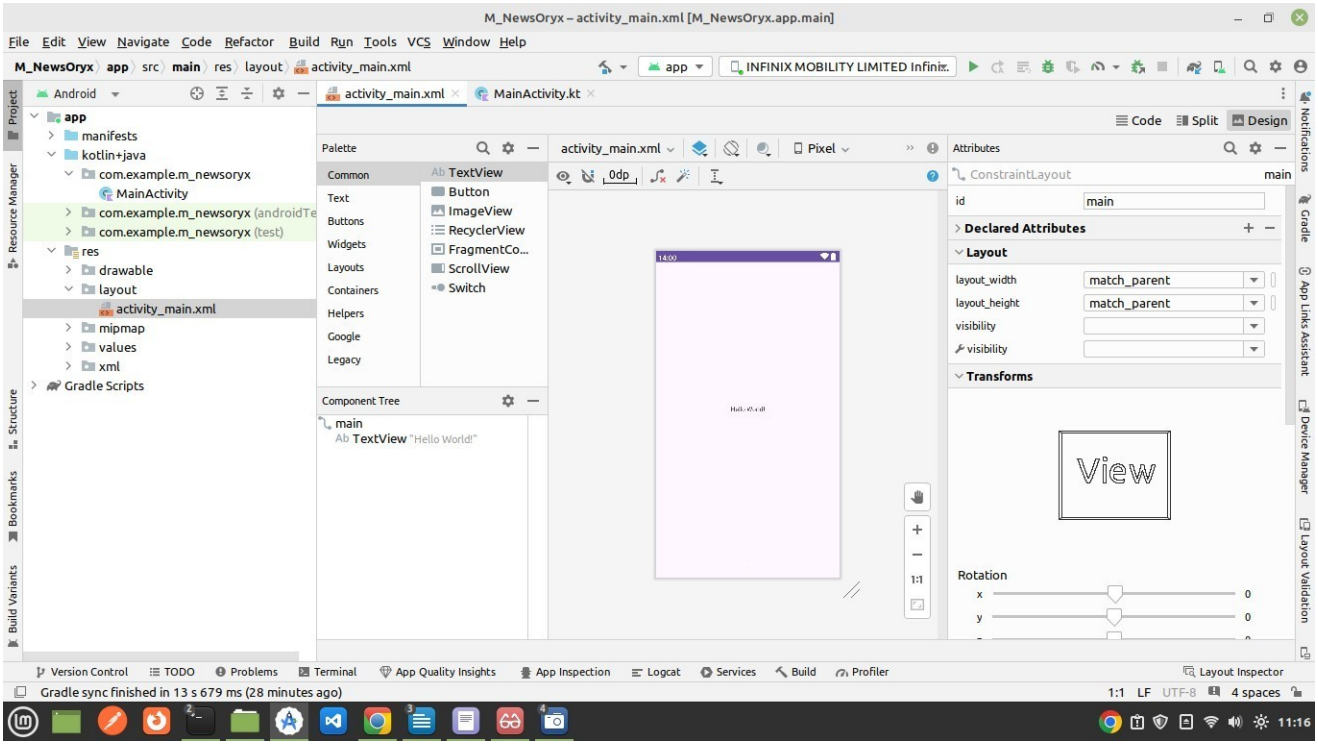
Drawable Folder (res/drawable)

It will contain the different types of images as per the requirement of application. It’s a best practice to add all the images in a **drawable** folder.



Layout Folder (res/layout)

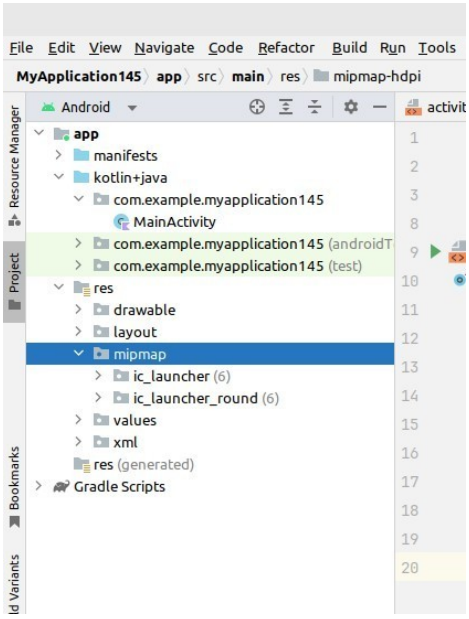
This folder will contain all XML layout files which we used to define the user interface of our application. Following is the structure of the **layout** folder in the android application.



Mipmap Folder (res/mipmap)

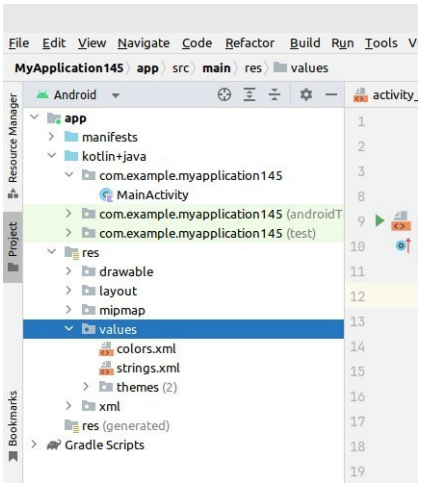
This folder will contain app / launcher icons that are used to show on the home screen. It will contain different density type of icons such as hdpi, mdpi, xhdpi, xxhdpi, xxxhdpi, to use different icons based on the size of the device.

Following is the structure of the **mipmap** folder in the android application.



Values Folder (res/values)

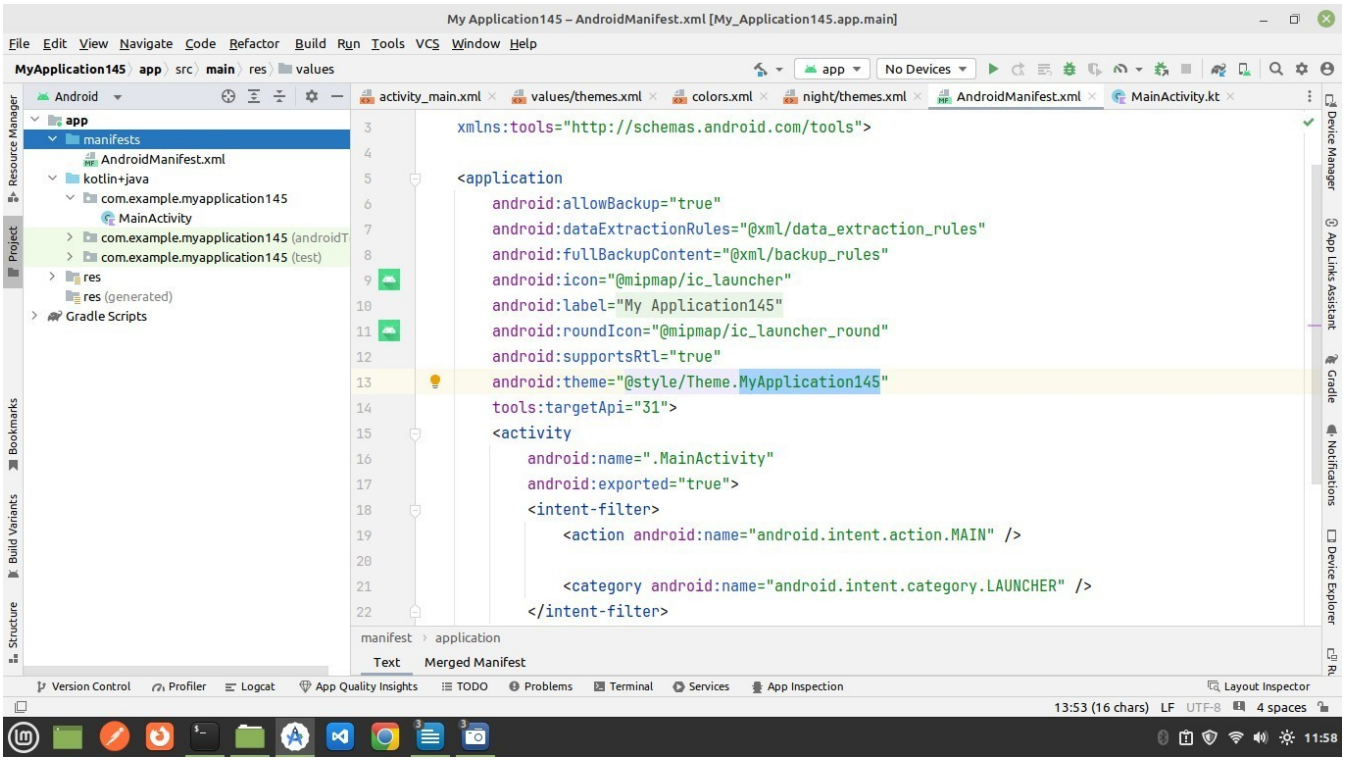
This folder will contain various XML files, such as strings, colors, strings and themes. Following is the structure of the **values** folder in android application.



Manifests Folder

This folder will contain a manifest file (**AndroidManifest.xml**) for our android application. This manifest file will contain information about our application such as android version, access permissions, Activities etc. of our application and its components. The manifest file will act as an intermediate between android OS and our application.

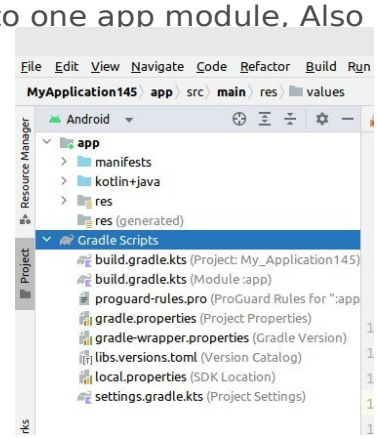
Following is the structure of the **manifests** folder in the android application.



Gradle Scripts

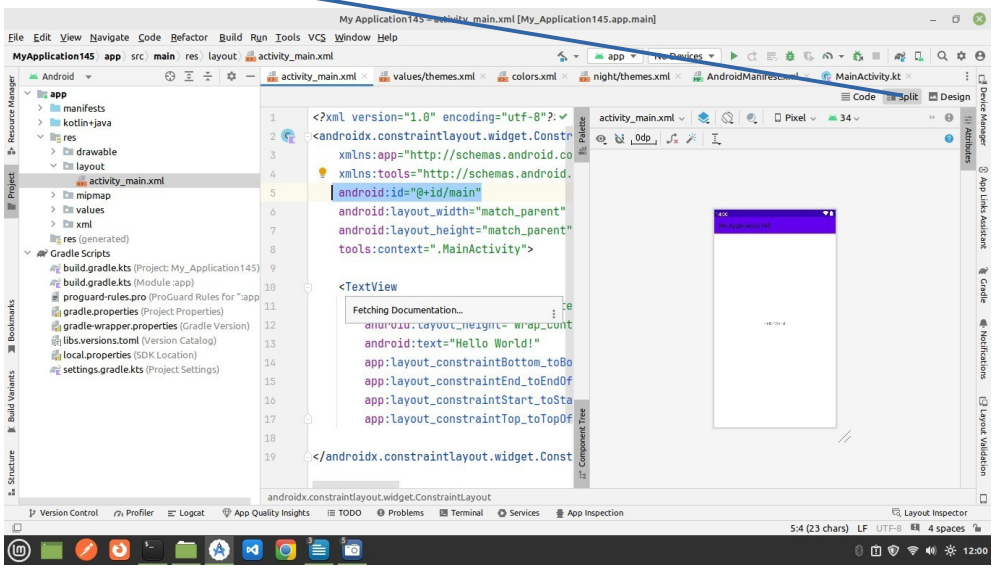
In android, Gradle means automated build system and by using this we can define a build configuration that applies to all modules in our application. In Gradle **build.gradle (Project)**, and **build.gradle (Module)** files are useful to build configurations that apply to all our app modules or specific to one app module. Also app dependencies are
found in these files.

Following is the structure of **Gradle Scripts** in the android application.



Android Layout File (activity_main.xml)

The UI of our application will be designed in this file and it will contain **Code, Split** and **Design** modes. The layout exists in the **layouts** folder and the structure of **activity_main.xml** file in **Split** mode looks like as shown below.



Go To res Folder , then Layout Folder open the *activity_main.xml*

Layout Explanations – Understanding the Layout Sections

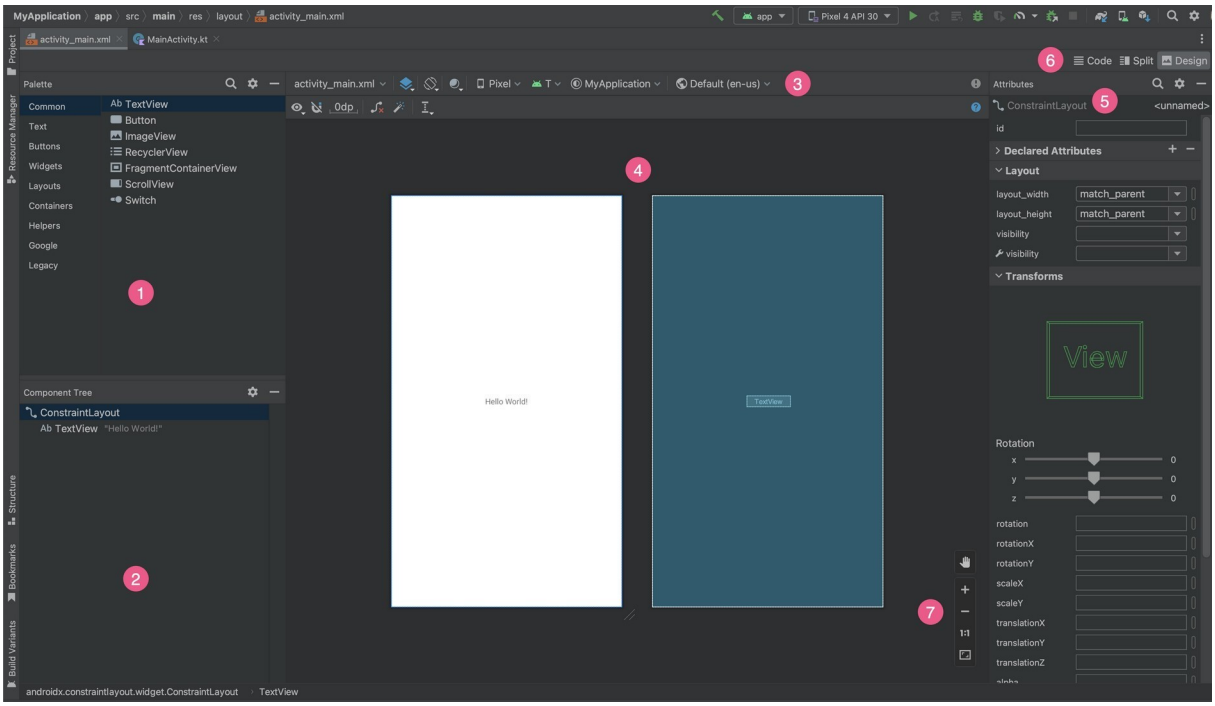


Figure 1. *The Layout Editor.*

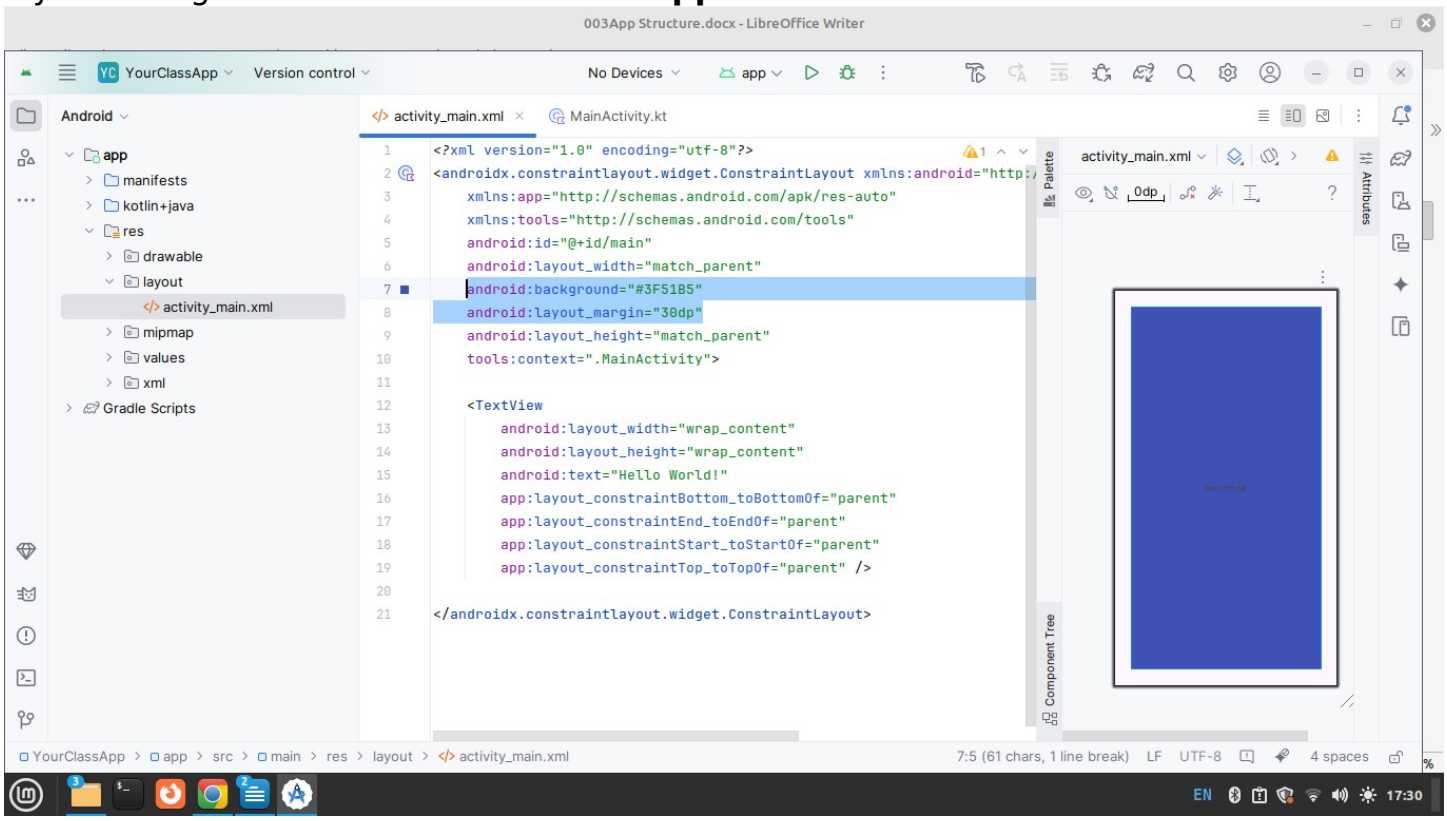
1. **Palette**: contains various views and view groups that you can drag into your layout.
2. **Component Tree**: shows the hierarchy of components in your layout.
3. **Toolbar**: has buttons that configure your layout appearance in the editor and change layout attributes.
4. **Design editor**: lets you edit your layout in Design view, Blueprint view, or both.
5. **Attributes**: has controls for the selected view's attributes.
6. **View mode**: lets you view your layout in either **Code**, **Split**, or **Design** modes. **Split** mode shows the **Code** and **Design** windows at the same time.
7. **Zoom and pan controls**: control the preview size and position within the editor.

When you open an XML layout file, the design editor opens by default, as shown in figure 1. To edit the layout XML in the text editor, click the **Code** button in the top-right corner of the window. Note that the **Palette**, **Component Tree**, and **Attributes** panels are not available while editing your layout in **Code** view.

Tip: To switch between design and text editors, press `Alt` (`Control` on macOS) plus `Shift` and the right or left arrow key.

Next, Go to *res - layout – activity_main.xml*

Below shows and android XML file, shows a layout, For these lesson, we change the layout background color. **And Run the App in our Phones**



Read more here
https://developer.android.com/studio/write/layout-editor